

IAN MBURU

BSc Computer Technology | IT Support • Networks • Distributed Systems

Nairobi, Kenya • kanyoroian@gmail.com • linkedin.com/in/ian-mburu-19a388240 • github.com/fo401 • fo401.github.io

PROFILE

Computer Technology graduate working across both ends of a systems stack: frontline IT and network support in a high-traffic public service environment, and full-stack distributed software built for settings where connectivity and infrastructure cannot be relied on. Designed and delivered a permissioned blockchain platform spanning two organisations — Hyperledger Fabric ledger, containerised deployment, role-based React dashboards, and 87/87 backend tests passing against a live network. Open to roles in IT support, systems and network administration, data, and AI-adjacent engineering.

TECHNICAL SKILLS

Networking & infrastructure: LAN/WAN troubleshooting, network administration, structured diagnostics, virtual desktop infrastructure (VDI), operating system deployment, endpoint and printer administration, hardware maintenance

Development: Node.js 20, Express, JavaScript (ES6+), React 18, Vite, HTML, CSS, Tailwind CSS, REST API design

Data & storage: MySQL 8, CouchDB, schema design, migrations and seeding, data integrity, backup and recovery

Distributed systems: Hyperledger Fabric v2.5 (RAFT ordering, chaincode, fabric-ca), MQTT (Mosquitto), Socket.io, event-driven architecture, IoT telemetry

Security: JWT (RS256) authentication, role-based access control, bcrypt password hashing, input validation, rate limiting, parameterised SQL, OWASP ZAP scanning, information security awareness

Testing & QA: Jest, Supertest, JMeter load testing, user acceptance testing, data-integrity and tamper testing

DevOps & tooling: Docker, Docker Compose, Git, GitHub, GitHub Pages, Linux / WSL2, npm

Practice: Technical documentation, incident triage and escalation, reporting, recurring security auditing

PROJECTS

Decentralized Cold Chain Vaccine Monitoring and Alerting System

Final-year research project

[Hyperledger Fabric v2.5](#) • [Node.js](#) • [Express](#) • [React](#) • [MySQL](#) • [CouchDB](#) • [Docker](#) • [MQTT](#) • [Socket.io](#) • [Africa's Talking SMS API](#)

github.com/fo401/cold-chain-system

- Designed and built a permissioned two-organisation Fabric network (Ministry of Health and KEMSA) recording every sensor reading to a shared ledger, making vaccine temperature history jointly auditable rather than self-reported by whoever holds the stock.
- Engineered asymmetric alert escalation: heat breaches above +8 °C escalate sequentially through four tiers on WHO time windows, while freeze breaches below −0.5 °C broadcast simultaneously to all roles, because potency loss from freezing is irreversible within the hour.
- Routed offline-buffered readings replayed past the damage window straight to mandatory quarantine, writing every quarantine decision to an immutable six-step chain of custody.
- Delivered scope-filtered, real-time React dashboards over Socket.io for five operational roles, from Facility Custodian to National EPI Manager.
- Secured the API with JWT RS256, bcrypt, Helmet, express-validator and rate limiting, and held personally identifiable information off-chain in line with the Kenya Data Protection Act 2019.
- Verified against a live Fabric network: 13 chaincode functions and three escalation modes proven end to end, with 87/87 backend tests passing (Jest + Supertest).
- Containerised the full stack with Docker Compose, simulated ESP32 IoT telemetry over MQTT, and released the project under Apache 2.0.

Vellum Archive — The Athenaeum

Personal project

[Hand-authored HTML](#) • [CSS](#) • [JavaScript](#) • fo401.github.io/vellum

- Built and operate a live personal literary archive as a single hand-authored HTML file with no framework, build step, or third-party runtime dependency — a deliberately zero supply-chain attack surface.
- Maintain the site to a zero-defect standard through recurring multi-pass security and quality audits covering security headers, dependency surface, link integrity and content rendering.

EXPERIENCE

IT Support Intern — Huduma Centre, Mombasa

Jun – Sep 2024

ICT Division, State Department for Public Service

- Diagnosed and resolved network faults at both LAN and WAN level, maintaining connectivity across a high-traffic public service centre.
- Administered virtual desktop infrastructure (VDI) supporting day-to-day staff operations.
- Installed, configured and maintained operating systems, applications and software across staff workstations.

- Repaired and installed local and networked printers, and maintained data equipment and computer hardware.
- Provided frontline technical support to internal users, triaging and escalating faults to keep service counters operational during public hours.

EDUCATION

BSc Computer Technology — Multimedia University of Kenya

Sep 2022 – Nov 2026

Final examinations completed June 2026; graduating November 2026.

Relevant coursework: Computer Networks · Data Communication and Telecommunications · Data Structures and Algorithms · Database Systems · Operating Systems · Computer Architecture and Organisation · Digital Electronics · Microprocessor Systems · Embedded Systems · Distributed Systems · Information Security and Cryptography · Software Engineering · Object-Oriented Programming · Web Application Development · Research Methods

Member, MMU Computing and Information Technology (CIT) Club.

CERTIFICATIONS

Cisco Certified Network Associate (CCNA) — in progress

ISOC Network Professional — in progress

LANGUAGES

English — fluent (professional working proficiency) · Kiswahili — fluent (native)